

AMD DEVELOPER HACKATHON ACT II · TRACK 3 UNICORN

paperflow.

The document reconciler you
can point at confidential
paperwork, because nothing
has to leave your hardware.

● Local: AMD MI300X · Remote: Fireworks AI

open weights only · runs in a container

THE PROBLEM

Some paperwork legally cannot go to cloud AI.

Singapore banks process tens of thousands of KYC onboarding files a year, and MAS effectively bans sending those documents to any cloud AI. So the hardest part, reconciling identity documents that disagree across a pile, stays manual, slow, and queued at every compliance team in the country.

10,000s

onboarding files a year,
per bank

0

cloud document tools
cleared to touch them

100%

of the reconciliation done
by hand today

WHY THIS ISN'T ALREADY SOLVED

The two obvious answers are each half a product.

"REDACT BEFORE THE LLM"

A commodity.

Reversible redaction with consistent placeholders is already productised by Gravitee, Philter, TrueFoundry and LangChain, and free as MIT open source. But it is built for chat and agent prompts, not messy document piles.

"CLOUD IDP RECONCILIATION"

Runs in the wrong place.

Rossum, Infrd and Super.AI already match fields across documents. But the matching runs in the vendor's cloud, which is exactly what a privacy-blocked organisation cannot use.

Neither pillar alone is the wedge. The intersection is the product.

THE WEDGE

Data **minimisation**, not data residency.

DATA RESIDENCY · WHAT EVERYONE OFFERS

Bedrock, Vertex, Claude Desktop on third-party. "Bring the model to your cloud." You control *where* the endpoint runs.

documents → Farid Hassan → cloud model

The real identity still reaches the endpoint.

DATA MINIMISATION · PAPERFLOW

The real identity never reaches the reasoning model at all. Presidio tokenises before anything crosses.

documents → local map → [PERSON_1] →

cloud model

Only tokens cross. The map stays on your machine.

Relocating the model is commodity BYO-cloud. Removing the identity is the whole product.

WHO THIS IS FOR

Confidential piles that cannot leave the building.

Low-volume, high-sensitivity, heterogeneous document piles in organisations banned from sending documents to any cloud at all. The buyer is compliance, or the Data Protection Officer. The driver is a hard regulatory line, not a preference.

document-native

∩

reconciliation with rationale

∩

zero cloud egress

=

the cell nobody serves

HOW IT WORKS

Five agents. The sensitive work stays local.

Extract

local VLM · MI300X

Resolve + Redact

local · consistent map

Reconcile

Fireworks · redacted

Audit

local · gap-check

Emit

local · rehydrate

The twist: one consistent entity map across the whole pile.

Acme Corp, ACME Corporation and Acme Inc all resolve to `[ORG_1]` in every document. The cloud model reasons about relationships between placeholders without seeing a single real value, so redaction preserves reconciliation instead of breaking it.

THE PRODUCT

Trust is the interface, not a footnote.

- **Per-exchange privacy receipt**

Shown inline every turn, never behind a toggle. A routing chip states the route: "Local only · 0 cloud calls" or "Local + Cloud · 1 redacted call".

- **Pre-send review gate**

If a message would cross to the cloud with detected values, the send button arms to "Confirm · send redacted". Human consent before anything leaves.

- **One entity, one colour, three surfaces**

Underline on the real value in your message, a dashed token in the receipt, a solid chip in the record. Hover one and its matches light up across panes.

- **Detected-entities panel**

Every entity Presidio caught, with a "flag a missed one" control. Detection recall you can audit live, not a claim you have to trust.

the cloud saw only:

[PERSON_1]

[ADDRESS_2]

[ID_3]

[DOB_4]

[ORG_6]

The receipt is the interface.

THE ASSISTANT · RECONCILER · LIVE

Give me a summary of what was reconciled across this pile.

PRIVACY RECEIPT

● Local + Cloud · 1 redacted call

ROUTED HYBRID cross-document comparison · benefits from a larger remote model

[PERSON_1]

[ORG_2]

[ID_3]

[ADDR_4]

[DATE_5]

· 5 tokens crossed · 0 detected identifiers

Two clients identified across the four docs. **Mohammed Farid bin Hassan** (alias **Farid Hassan** merged automatically) has an address conflict between the KYC form and utility bill · MAS Notice 626 says the form is authoritative. **Yvonne Goh** has an NRIC OCR conflict on the scan · digit-form beats letter-substitution. Records are in the panel; every value cites its source.

- **Every turn shows the routing decision.**

A chip says which path the question took. Never behind a toggle.

- **The tokens above are what crossed to Fireworks.**

The count and the token list are the projection of the router's log · not a summary of it.

- **Air-gapped mode flips the chip to Local only · 0 cloud calls.**

The token list vanishes because the receipt says nothing crossed. Zero egress, provable.

THE LANDSCAPE

The empty cell is the combination.

CAPABILITY	PRIVACY LLM GATEWAYS	CLOUD IDP	PAPERFLOW
Messy multi-document piles	No	Yes	Yes
Cross-doc reconciliation with rationale	No	Partial	Yes
Per-field provenance	Limited	Partial	Yes
Reversible redaction round-trip	Yes	No	Yes
Runs with zero cloud egress	Yes	No	Yes
Gap-flagging against a checklist	No	Partial	Yes

We don't out-redact a gateway or out-scale a cloud IDP. We're the only viable option for a confidential pile that cannot leave.

MEANINGFUL USE OF AMD

The MI300X does the load-bearing confidential work.

GEMMA-4-31B-IT · 192 GB HBM3

Google's Gemma-4-31B-IT vision-language model, BF16 on a single MI300X via vLLM ROCm. Floor test 2026-07-07: 48/50 strict value recovery, ~5 s/page steady, sensitive extraction never leaves the box.

RAW-SENSITIVE, LOCAL

Local extraction and local entity resolution carry every raw step. The cloud only ever sees redacted tokens.

ZERO EGRESS

Air-gapped mode: one toggle, everything on the MI300X + container, zero egress by construction, for piles that cannot even send tokens.

Not a garnish. The sensitive work cannot happen anywhere but on local AMD hardware.

REALISED AND FUNCTIONAL

It runs end to end, and we score it.

Containerised from day one. A live run on a realistic pile, not slides. Four green eval suites against a known ground truth, printed with every commit:

Reconciliation

10 / 10

conflicts resolved · 1 escalated by design (owner review)

Gap-flagging

7 / 7

required fields missing across every pile flagged

Entity resolution

3 / 3

alias groups merged, not flagged as duplicates

Redaction recall

38 / 38

sensitive spans masked across the three demo piles

Plus receipts-vs-log consistency + prompt-injection safety, both green. `python -m eval.scorer` · reproducible on any fresh clone.

WHAT WE DON'T CLAIM

Sharply reduced exposure, **not** zero leakage.

Security rests on detection recall. Structured identifiers are high-recall; person and company names rely on NER and will miss edge cases. So the UI says "0 detected identifiers crossed", never "zero leakage". The pre-send gate and the detected-entities panel are that caveat rendered as features.

For un-patternable free text, **full-local mode is the honest answer**, and there the absolute claim is true by construction. We frame paperflow for environments where cloud document AI is off-limits, not as a certified compliance product.

WHERE IT GOES

Same engine, any pile that cannot leave.

FINANCE · LIVE DEMO

KYC / AML onboarding

The sector that banned cloud LLMs first and hardest.

HEALTHCARE

Patient intake

Records that disagree across referrals and forms.

ENTERPRISE

Partner collation

Confidential counterparties across a deal pile.

A tab switch in the demo, and the on-screen proof it generalises. Wherever cloud document AI is off-limits, paperflow is the reconciler that can still run.

paperflow.

Reconciliation for paperwork that legally
cannot leave the building.

On your own AMD hardware.

● AMD MI300X · Fireworks AI · Microsoft Presidio

original · MIT-licensed · open weights only